Student Number:

THE UNIVERSITY OF MELBOURNE

Department of Finance
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SAMPLE EXAM

SUBJECT NUMBER:	**FNCE30007**
SUBJECT NAME:	**Deriva ve Securi es**
EXAM DURATION:	**3 HOURS**
READING TIME:	**15 minutes**
TOTAL MARKS:	**75**

NOTE: IN ALL YOUR ANSWERS THAT REQUIRE CALCULATIONS, PLEASE SHOW YOUR COMPLETE WORK

QUESTION 1 [7 MARKS]

A 9-month American op on allows an Italian investor to purchase 1 million USD at the rate of 0.85 euros per USD. The current spot FX rate is 0.90 euros per US, the FX rate vola lity is 15% p.a., the risk-free rate in the US is 1% p.a. and in Europe it is 2.5% p.a.

a) Use a three-period binomial model to assess the total premium paid by the investor (round to 4 decimal places at each step)

S0	0.9
rf	0.01
sigma	0.15
u	1.0779
d	0.9277
rf	0.025
Dt	0.25
K	0.85
p	0.5064

Tree for the FX rate is

			1.1271
		1.0457	
	0.9701		0.9701
0.9		0.9	
	0.8349		0.8349
		0.7746	
			0.7186

Tree for the Op on value is

			0.2771
		0.1984	
	0.1295		0.1201
0.0801		0.0604	
	0.0304		0
		0	
			0

➔ Op on price is es mated to be 8.01 cents of 1 euro. Hence, Total Premium =0.0801*1 mil = 80,100 euros

b) What posi on does a financial intermediary wri ng the op on need to take in FX to delta-hedge its posi on?

From the trees above, the Delta of the op on at the current FX rate is

(0.1295 – 0.0304) / (0.9701 – 0.8349) = 0.7328

➔ Delta of op on writer is -0.7328*1,000,000 = -732,800

➔ Op on writer needs to take long posi on of USD 732,800 to delta-hedge the op ons

QUESTION 2 [5 marks]

A stock price is currently \$30. During each two-month period for the next four months it is expected to increase by 8% or reduce by 10%. The risk-free interest rate is 5%.

a) Use a two-step tree to calculate the value of a derivative that pays off $\max[(30-S_T),0]^2$ where S_T is the stock price in four month.

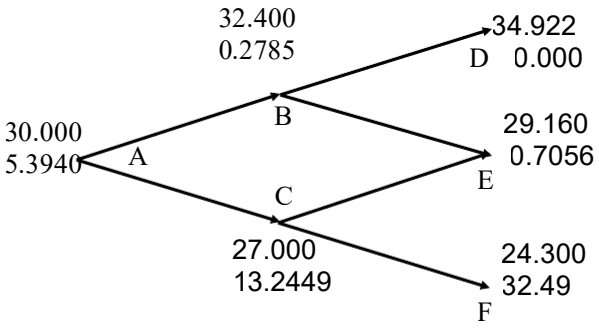
b) If the derivative is American-style, should it be exercised early?

(Round to 4 decimal places at each step)

The risk-neutral probability of an up move, p , is given by

$$p = \frac{e^{0.05\times 2/12} - 0.9}{1.08 - 0.9} = 0.6020$$

The value of the European option is 5.394. This is calculated by working back through the tree shown in the Figure below. The second number at each node is the value of the European option.
Early exercise at node C would give 9.0 which is less than 13.2449. The option should therefore not be exercised early if it is American.



QUESTION 3 [6 marks]

A company wishes to hedge its exposure to a new fuel whose price changes have a 0.6 correlation with gasoline futures price changes. The company will lose $1 million for each 1 cent increase in the price per gallon of the new fuel over the next three months. The new fuel's price change has a standard deviation that is 50% greater than price changes in gasoline futures prices. The company wants to implement a hedging strategy that minimizes its overall risk, measured as the variance of the hedged position

a. What is the company's exposure measured in gallons of the new fuel?
b. If gasoline futures (whose contact size is 42,000 gallons) are used to hedge the exposure what should company do to achieve its hedging goals? Specifically:
    i. What position measured in gallons should the company take in gasoline futures?
    j. How many gasoline futures contracts should be traded?

Define X as the unknown exposure, measured in gallons of new fuel.
Then, X*$0.01 = $1,000,000
➔ X = 1,000,000/0.01 = 100,000,000
The company has, thus, an exposure to the price of 100 million gallons of the new fuel.

The variance minimizing hedge ratio is 0.6 × 1.5 = 0.9.

The company should, therefore, take a position of 100,000,000*0.9 = 90 million gallons in gasoline futures.

The number of contracts required is therefore

$$\frac{90{,}000{,}000}{42{,}000} = 2142.9$$

or, rounding to the nearest whole number, 2143.

Ques on 4 [6 Marks]

It is now October 2013. A company anticipates that it will purchase 1 million pounds of copper in February 2014 and August 2014. The company has decided to use the futures contracts traded in the COMEX division of the CME Group to hedge its risk. One contract is for the delivery of 25,000 pounds of copper. The initial margin is $2,000 per contract and the maintenance margin is $1,500 per contract. The company's policy is to hedge 80% of its exposure. Contracts with maturities up to 13 months into the future are considered to have sufficient liquidity to meet the company's needs. Assume the market prices (in cents per pound) today and at future dates are as follows.

Date	*Oct 2013*	*Feb 2014*	*Aug 2014*
Spot Price	*372.00*	*369.00*	*365.00*
Mar 2014 futures price	*372.30*	*369.10*	
Sept 2014 futures price	*372.80*	*370.20*	*364.80*

Devise a hedging strategy for the company.

To hedge the February 2014 purchase the company should take a long position in March 2014 contracts for the delivery of 800,000 pounds of copper. The total number of contracts required is $800,000 / 25,000 = 32$. Similarly a long position in 32 September 2014 contracts is required to hedge the August 2014 purchase.

The strategy is therefore as follows
Oct 2013: Enter into long position in 32 Sept. 2014 contracts
Enter into a long position in 32 Mar. 2014 contracts
Feb 2014: Close out 32 Mar. 2014 contracts
Aug 2014: Close out 32 Sept. 2014 contracts

What is the impact of the strategy you propose on the price the company actually pays for copper in Feb 2014?

With the market prices shown the company pays

$$369.00 + 0.8 \times (372.30 - 369.10) = 371.56$$

for copper in February, 2014.

QUESTION 5 [2+2+2+2 = 8 marks]

Consider an American put option on a non-dividend paying stock when the stock price is $30, the strike price is $32, the risk-free rate is 5%, the volatility is 30%, and the time to maturity is 1.5 years.

a. What is the option's intrinsic value?
b. *Without* computing the option premium, do you expect such premium to be higher or lower than the intrinsic value? Explain your answer
c. What would a time value of zero indicate? What is the value of an option with zero time value?
d. Consider now a new put option (not the one above) with negligible time value and positive intrinsic value. Would such option more likely be deep-in-the-money or deep-out-of-the-money? Explain your answer

a) The option's intrinsic value is $32-30=2.00$.
b) Since the put is in-the-money, its value certainly exceeds the intrinsic value
c) A time value of zero would indicate that it is optimal to exercise the option immediately. In this case the value of the option would equal its intrinsic value.
d) Deep-in-the-money. To have positive intrinsic value the option must be in the money. Moreover, when they are deep in the money (look like stocks) heir time value is very small.

QUESTION 6 [6 marks]

Carefully explain the concept of implied vola lity and its importance for op on pricing and, more generally, for market par cipants. Make sure you illustrate the VIX index.

Implied volatility for an option is the value of the volatility of the underlying asset returns (or, prices) that makes the observed market price of the option equal to the price computed according to a specific option pricing model, such as the Black-Scholes-Merton model.

It is considered a good indicator of the future volatility of the underlying asset as it summarizes the opinion of market participants as reflected in the traded option prices. As such, it is seen as a forward-looking estimate (or, forecast) of volatility

The most well-known implied volatility measure is the VIX Index for S&P500 options traded on CBOE. It is computed as an average of short-maturity, mostly ATM index options on the S&P500. It is seen as a barometer of investors' opinion about the level of uncertainty (or, "fear") in the market place. The VIX typically goes up in periods of market stress and, hence, greater uncertainty about where market values are headed

----------YOUR ANSWER MUST END HERE OR ABOVE----------

Ques on 7 [8 Marks]

A non-dividend-paying stock is currently trading at $32, its volatility is 30%, and the risk-free rate for all maturities is 5% per annum. Mr Peterman believes the stock is overpriced an decides to set up a bear spread using European put options with strike prices of $25 and $30 and a maturity of six months. Mr Peterman believes that the market price of the put options will closely match the BSM model price.

Assuming Mr Peterman will turn out to be correct about the put premiums, what must happen to the stock price for Mr Peterman's strategy to be profitable? (round to 4 decimal digits)

Using the BSM model, a put option with a strike price of 25 costs 0.2796 and a put option with a strike price of 30 costs 1.4430. The cost of the bear spread is therefore $1.4430 - 0.2794 = 1.1636$. The profits ignoring the impact of discounting are

Stock Price Range	*Profit*
$S_T \leq 25$	$30 - 25 - 1.1636 = +3.8364$
$25 < S_T < 30$	$28.8364 - S_T$
$S_T \geq 30$	-1.1636

➔ **For Mr Peterman to turn a pro t (before transac ons costs) the stock price must drop below $28.8364 ($28.84 in actuality)**

QUESTION 8 [15 marks]

Costanza Bank, located in Melbourne, Australia, has the following portfolio of over-the-counter European options on the British Pound (symbol £):

Type	Position	Delta of Option	Gamma of Option	Vega of Option
Call	-1,000	0.50	2.20	1.80
Call	-500	0.80	0.60	0.20
Put	-2,000	-0.40	1.30	0.70
Call	-500	0.70	1.80	1.40

As hedging instruments the bank is considering traded European options on the £. The spot FX rate is currently 1.80 Australian Dollars (AUD) for 1 £, with a volatility of 15% p.a. The risk-free interest rate is 1% in Great Britain and 2.5% in Australia.

Specifically, they are looking at 7-month ATM (i.e., K=1.80 AUD for 1 £) calls.

a. What should the bank do to make the portfolio both gamma neutral and delta neutral?

b. What should the bank do to make the portfolio both vega neutral and delta neutral?

c. Assume the bank decides to set up the delta-gamma neutral position in a. After a month, the spot FX rate moves to 1.90 AUD per £, while its volatility stays the same. Risk-free rates of interest also stay the same. The deltas of the over-the counter options change and become as follows

Type	Position	Delta of Option
Call	-1,000	0.72
Call	-500	0.86
Put	-2,000	-0.53
Call	-500	0.88

Facing this situation, the bank decides to keep only delta neutrality: what does the bank need to do in order to accomplish such goal?

For all answers show your calculations.

a.

The delta of the unhedged portfolio is

$$-1,000\times 0.50-500\times 0.80-2,000\times(-0.40)-500\times 0.70=-450$$

The gamma of the unhedged portfolio is

$$-1,000\times 2.2-500\times 0.6-2,000\times 1.3-500\times 1.8=-6000$$

The vega of the unhedged portfolio is

$$-1,000\times 1.8-500\times 0.2-2,000\times 0.7-500\times 1.4=-4,000$$

If the bank wants to use options with the above specs, it first needs to compute their Greeks. For those options:

S= 1.80 K = 1.80 sigma = 15% r=2.5% r_foreign = 1% T=7/12

Plugging these inputs into the Greek formulas one gets:

Delta=0.5499
Gamma = 1.9062
Vega = 0.5404

To make the portfolio gamma neutral a long position in +6,000/1.9062 = 3147 (or 3148) traded £ options needs to be taken

The delta of the gamma neutral portfolio is then: =3147*0.5499 -450 = +1280.54

Hence, in addition to the 3147 traded options, a short position of 1280 in £ is necessary so that the portfolio is both gamma and delta neutral.

b. A long position in +4,000/ 0.5404 = 7402 traded options would give a vega-neutral portfolio

The delta of the vega-neutral portfolio is then 7402*0.5499-450 = 3630.36

Hence, in addition to the 7402 traded options, a short position of 3630 in £ is necessary so that the portfolio is both vega and delta neutral.

c. At the new spot FX rate of 1.90 AUD/£ the Delta of the traded options (which now have T=6/12) is 0.7331.

Hence, the delta of the overall (options + spot sterling) portfolio is

-1000*0.72 -500*0.86 -2000*-0.53 -500*0.88 + 3147*0.7331 -1280*1 = +497

To re-establish a delta neutral portfolio the bank need to short an additional 497 in pounds

Question 9 [3 Marks]

It is 10:00 AM AEDT on a Tuesday morning and the share price of NAB is $29. NAB has announced that it will pay a $0.50 dividend in two months and a dividend of the same amount in five months. The risk-free rate of interest is 10% per annum. You observe that a six-month European call option on NAB with a strike of $30 just traded for $2. A 6-month European put on NAB with a strike of $30 has not traded yet during the day.

At what price should you expect to trade such put at 10:00 AM AEDT (same morning) assuming arbitrage opportunities will not arise?

When the underlying asset pays income, the put-call parity for European options is

$$p + S = c + K*\exp(-rT) + D$$

where D s the PV of the dividends paid during the life of the option

So, the no-arbitrage price of the put must be:
$p = 2 + 30*\exp[(-0.10)(6/12)] + (0.5*\exp[(-0.10)(2/12)] + 0.5*\exp[(-0.10)(5/12)] - 29 =$ $2.51

Question 10 (4 Marks)

In the CNBC video on options posted in Week 11, the trader suggest to implement a trading strategy using VIX options. *Briefly* describe the details of strategy and its purposes. As you do so, please indicate what is the basic market dynamics the strategy seeks to exploit.

The suggested strategy is a bull spread using call options on the VIX expiring in 1 month (from the time the video was recorded)

It is recommended as a hedging strategy against a possible drop in the US equity market

It aims to exploit the negative correlation between the level of the US equity market (measured by the S&P500 index) and the level the VIX

----------YOUR ANSWER MUST END HERE OR ABOVE----------

Question 11 (3+2+2=7 Marks)

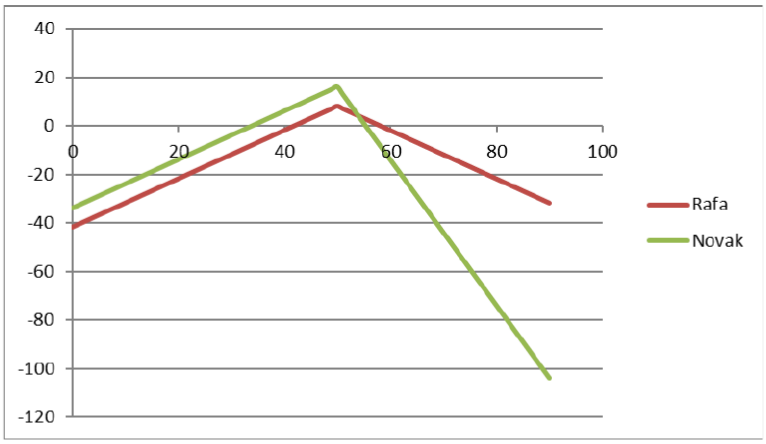
Rafa and Novak both expect that stock price of eBay, currently at $50, will not move substantially over the next 2 months. They both decide to purchase the stock and write 2-month at-the-money call options on the stock. They disagree only on the amount of options to write. Specifically, their positions are as follows:

RAFA: Long one share and short two calls
NOVAK: Long one share and short four calls

The call options are trading at $4.05

a) Draw a diagram showing the variation of each investor's profit and loss with the stock price at the expiration of the option.
b) Are Rafa and Novak implementing strategies that are consistent with their expectations?
c) Comment on the relative appeal of the two strategies if, contrary to what Rafa and Novak predict, eBay's stock price experience a large upward movement in the coming two months

a) The variation of the investor's profit/loss with the terminal stock price for each of the two strategies is shown in the figure below. The x-axis contains values of eBay stock price in 2 months. The y-axis measures total *net* profits.



b) The strategies are consistent with the expectations as both profit if, in 2 months, the stock price of eBay does not move substantially from current levels.
c) A sharp increase in the stock price would cause larger losses for Novak than for Rafa. Rafa's strategy is less exposed on the upside.